```
!pip install pgmpy
```

```
Collecting pgmpy
  Downloading pgmpy-1.1.0-py3-none-any.whl.metadata (13 kB)
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Installing collected packages: scikit-base, pgmpy
Successfully installed pgmpy-1.1.0 scikit-base-0.13.1
```

```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.inference import VariableElimination
import pandas as pd

# Step 1: Define Dataset
data = pd.DataFrame({
    'Rain': ['No', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No'],
    'TrafficJam': ['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No', 'No'],
    'ArriveLate': ['Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No']
})

# Step 2: Define Bayesian Network Structure (DAG)
model = DiscreteBayesianNetwork([
    ('Rain', 'TrafficJam'),
    ('TrafficJam', 'ArriveLate')
])

# Step 3: Train Model using MLE
model.fit(data)

# Step 4: Display CPDs
print("\n--- Conditional Probability Distributions (CPDs) ---")
for cpd in model.get_cpds():
    print(cpd)

# Step 5: Inference using Variable Elimination
inference = VariableElimination(model)
```

```
# Step 6: Query
result = inference.query(variables=['ArriveLate'], evidence={'Rain': 'Yes'})
print("\n--- Inference Result ---")
print(result)
```

```
/usr/local/lib/python3.12/dist-packages/pgmpy/estimators/__init__.py:4: FutureWarning: `pgmpy.estimators.StructureScore` is depr
  from .StructureScore import (

--- Conditional Probability Distributions (CPDs) ---
+-----------+-----+
| Rain(No)  | 0.5 |
+-----------+-----+
| Rain(Yes) | 0.5 |
+-----------+-----+
+-----------------+----------+-----------+
| Rain            | Rain(No) | Rain(Yes) |
+-----------------+----------+-----------+
| TrafficJam(No)  | 0.5      | 0.5       |
+-----------------+----------+-----------+
| TrafficJam(Yes) | 0.5      | 0.5       |
+-----------------+----------+-----------+
+-----------------+----------------+-----------------+
| TrafficJam      | TrafficJam(No) | TrafficJam(Yes) |
+-----------------+----------------+-----------------+
| ArriveLate(No)  | 0.75           | 0.25            |
+-----------------+----------------+-----------------+
| ArriveLate(Yes) | 0.25           | 0.75            |
+-----------------+----------------+-----------------+

--- Inference Result ---
+-----------------+-------------------+
| ArriveLate      |   phi(ArriveLate) |
+=================+===================+
| ArriveLate(No)  |            0.5000 |
+-----------------+-------------------+
| ArriveLate(Yes) |            0.5000 |
+-----------------+-------------------+
```